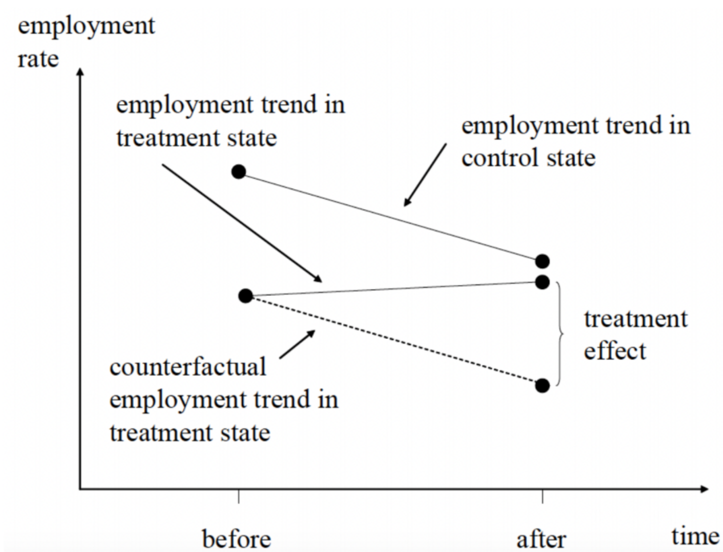
1. (a) This is one of the first applications of difference-in-differences (DID).

- New Jersey (NJ) raised the state minimum wage from \$4.25 to \$5.05 in April, 1992. The minimum wage in the neighbouring state Pennsylvania (PA) stayed at \$4.25.
- Card and Krueger collected data on employment at fast food restaurants in NJ in Feb 1992 (before the policy change) and again in Nov 1992 (after the policy change). They collected data from the same type of restaurants in eastern PA.
- They compared the Feb-to-Nov change in employment in NJ to the change in employment in PA over the same period to obtain the DID estimate of the effect of minimum wage change on employment.
- See the figure below for the graphical illustration. Note that to obtain the treatment effect, we need to assume that without the policy change (the raise in minimum wage), the NJ trend (treatment trend) would have moved in the same way as the PA trend (control trend). This is the key assumption of the DID design, which is called the **"parallel trend assumption"**.



Then, the DID estimate of the policy effect can be written as "Treatment Trend" - "Control Trend":

$$DID = \left[\bar{Y}_{After,NJ} - \bar{Y}_{Before,NJ}\right] - \left[\bar{Y}_{After,PA} - \bar{Y}_{Before,PA}\right] \quad (1)$$
$$= (20.90 - 20.43) - (21.10 - 23.38) = 2.75, \quad (2)$$

where $\bar{Y}_{After,NJ}$ denotes the average employment (in fast-food restaurants) in NJ after the policy change, etc.

Therefore, Card and Krueger's DID analysis shows that the employment in NJ was not negatively affected by the raise of the minimum wage. Their empirical result is in contrast to the prediction from the standard economic theory.

	New Jersey	Pennsylvania	Difference
Before Increase	20.43	23.38	2.95
After Increase	20.90	21.10	0.20
Difference	0.47	−2.28	**2.75**

Average employment at fast food restaurants in NJ and PA

(b) The linear-log regression model (with a single regressor) can be written as:

$$Y_i = \beta_0 + \beta_1 log(X_i) + u_i, i = 1, ..., n.$$

where $log(X_i)$ is the natural logarithm of X_i. The only difference between this model and the standard linear regression model is that the right-hand-side variable is now $log(X_i)$ rather than X_i itself.

The meaning of β_1 is as follows:

$$\beta_1 \approx \frac{\Delta Y}{\Delta X/X} = \frac{\text{change in Y}}{\text{percentage change in X}}.$$

For example, suppose $X = 100$ and $\Delta X = 1$, then $\Delta X/X = 1\%$. Suppose $\beta_1 = 2$, then the corresponding change in Y is: $\Delta Y \approx \beta_1 \times \Delta X/X = 2 \times 1\% = 0.02$ (unit of Y). Similarly, suppose $\Delta X/X = 10\%$ and $\beta_1 = 2$, then the corresponding change in Y is: $\Delta Y \approx \beta_1 \times \Delta X/X = 2 \times 10\% = 0.2$ (unit of Y).

2. (a) The difference-in-differences (DID) model can be defined as follows:

$$score = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 D_1 \times D_2 + u, \tag{3}$$

where D_1 and D_2 are two dummy variables. Specifically, $D_1 = 1$ for Western (treatment group), $D_1 = 0$ for Eastern (control group), $D_2 = 1$ for the time period after the treatment (2022), and $D_2 = 0$ for the time period before the treatment (2020). Then, we have totally four scenarios:

$$\{D_1 = 0,\ D_2 = 0\} = \{\text{Eastern, Before Treatment (fund increase)}\} \tag{4}$$
$$\{D_1 = 1,\ D_2 = 0\} = \{\text{Western, Before Treatment (fund increase)}\} \tag{5}$$
$$\{D_1 = 0,\ D_2 = 1\} = \{\text{Eastern, After Treatment (fund increase)}\} \tag{6}$$
$$\{D_1 = 1,\ D_2 = 1\} = \{\text{Western, After Treatment (fund increase)}\} \tag{7}$$

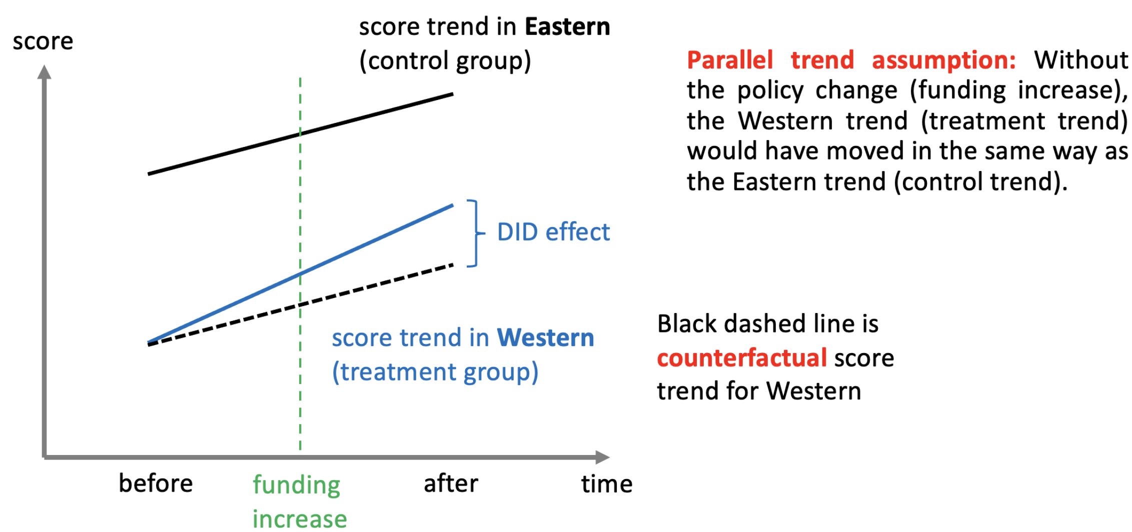
It is also clear that β_3 in (3), **the parameter for the interaction between D_1 and**

D_2, corresponds to the DID treatment effect:

$$\begin{aligned}\beta_3 &= [\{\text{Western, After Treatment}\} - \{\text{Eastern, After Treatment}\}] \\ &\quad - [\{\text{Western, Before Treatment}\} - \{\text{Eastern, Before Treatment}\}] \\ &= [\{\text{Western, After Treatment}\} - \{\text{Western, Before Treatment}\}] \\ &\quad - [\{\text{Eastern, After Treatment}\} - \{\text{Eastern, Before Treatment}\}] .\end{aligned}$$

Therefore, β_3 is the key parameter of interest in the model: Whether the additional funding for Western led to improved test scores is captured by β_3.

(b) Refer to the figure below (assuming that the Eastern trend is higher than the Western trend). The key assumption is the parallel trend assumption, which assumes that without the policy change (funding increase), the Western trend (treatment trend) would have moved in the same way as the Eastern trend (control trend).

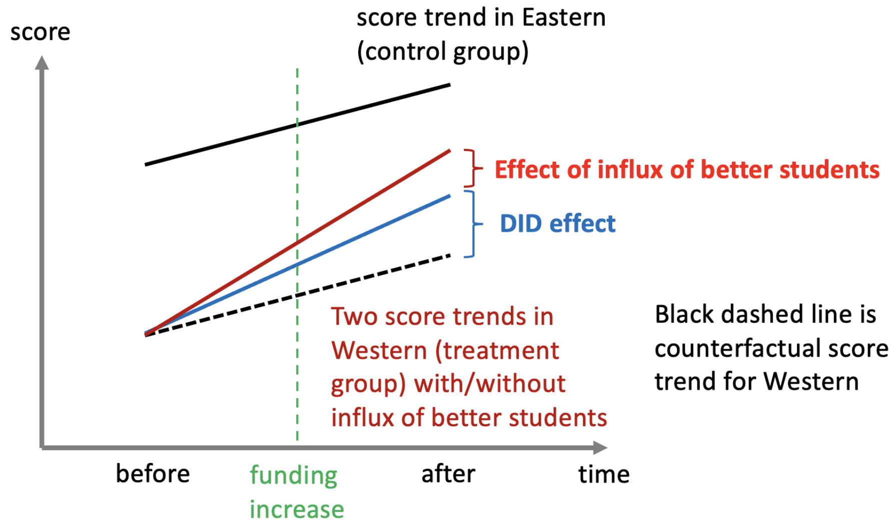


In the figure, we suppose the Eastern trend is higher than the Western trend.

(c) According to the figure below, the influx of better students to Western may induce an upward bias for the DID estimator. This can also be seen as follows:

$$\begin{aligned}&DID\ estimator \\ = \ &\left(\overline{score}_{W,After} - \overline{score}_{W,Before}\right) - \left(\overline{score}_{E,After} - \overline{score}_{E,Before}\right), \qquad (8)\end{aligned}$$

where $\overline{score}_{W,After}$ denote the average score of Western after the treatment (additional funding), etc. Then, we expect that $\overline{score}_{W,After}$ will increase due to the fund increase, so that the value of the DID estimator will be higher than it should be. Intuitively, **there are now two factors driving the DID estimator**: (a) the effect of the fund increase for Western (this is the treatment effect we are really interested in), and (b) the effect of the better students to Western. This results in **an upward bias** for the DID estimator.



3. (a) For the model $Y = \beta_0 + \beta_1 Treat + \beta_2 Treat \cdot After + u$, we have

$$\begin{aligned} E[Y|Treat = 0, After = 0/1] &= \beta_0, \\ E[Y|Treat = 1, After = 0] &= \beta_0 + \beta_1, \\ E[Y|Treat = 1, After = 1] &= \beta_0 + \beta_1 + \beta_2. \end{aligned}$$

Therefore, β_0 is the average outcome of the control group (for both before and after). **In a graphical illustration, the trend of the control group should be totally horizontal.**

In addition, we have

$$\beta_1 = E[Y|Treat = 1, After = 0] - E[Y|Treat = 0, After = 0/1],$$

which is the difference in average outcomes between the treatment group before treatment and the control group;

$$\beta_2 = E[Y|Treat = 1, After = 1] - E[Y|Treat = 1, After = 0],$$

which is the difference in average outcomes of the treatment group before and after the treatment.

(b) Similarly, for the model $Y = \beta_0 + \beta_1 After + \beta_2 Treat \cdot After + u$, we have

$$\begin{aligned} E[Y|Treat = 0/1, After = 0] &= \beta_0, \\ E[Y|Treat = 0, After = 1] &= \beta_0 + \beta_1, \\ E[Y|Treat = 1, After = 1] &= \beta_0 + \beta_1 + \beta_2. \end{aligned}$$

Therefore, β_0 is the average outcome of both treatment and control groups before the treatment. **In a graphical illustration, the trends of both treatment and control groups should start from the same point.**

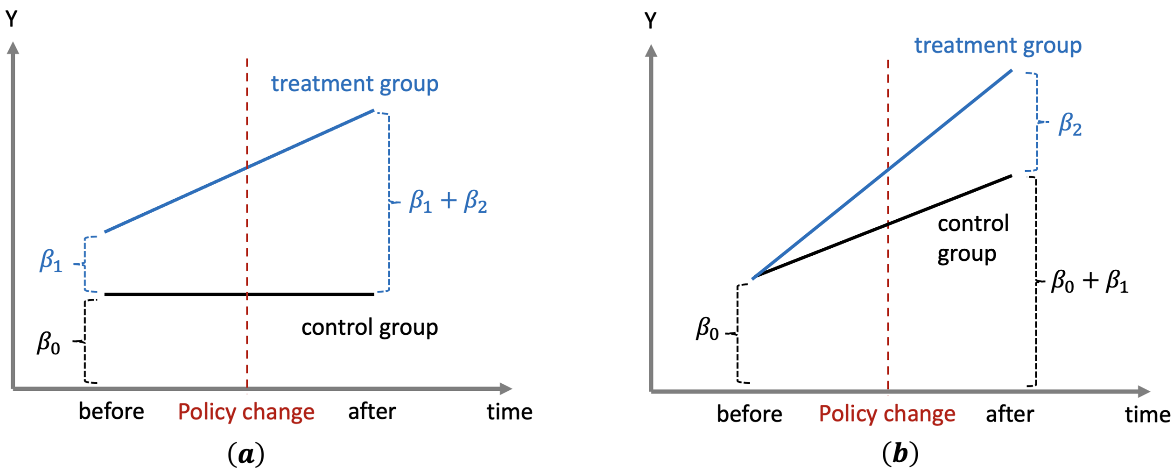
Furthermore, we have

$$\beta_1 = E[Y|Treat = 0, After = 1] - E[Y|Treat = 0/1, After = 0],$$

which is the difference in average outcomes between the control group after the treatment and the treatment/control groups before treatment;

$$\beta_2 = E[Y|Treat = 1, After = 1] - E[Y|Treat = 0, After = 1],$$

which is the difference in average outcomes between the treatment and control groups after the treatment.



(c) For a correct difference-in-differences (DID) model, the meaning of the coefficient of the interaction term is:

$$\begin{aligned}&[E[Y|Treat = 1, After = 1] - E[Y|Treat = 1, After = 0]]\\ &- [E[Y|Treat = 0, After = 1] - E[Y|Treat = 0, After = 0]]\,,\end{aligned}$$

which is equal to "difference of the treatment group before and after the treatment" - "difference of the control group before and after the treatment" (note that this is equivalent to "difference between two groups after the treatment" - "difference between two groups before the treatment").

Therefore, neither of the specifications in (a) and (b) is correct to obtain the parameter of interest. In particular, the model in (a) does not allow for the change in outcome of the control group before and after the treatment, while the model in (b) does not allow for the difference in outcomes between the two groups before the treatment. All the dummy variables $Treat$, $After$, and $Treat \cdot After$ are necessary for the correct specification of DID model.

4. (a) According to AJR's argument, the colonies where the early European settlers' mortality rates had been low were more likely to become the "migrant colonies" (some scholars

also called them "Neo-Europe"). By contrast, the colonies where the mortality rate had been high were more likely to become the "extractive states".

Therefore, in Panel B, the left-hand-side countries such as "NZL", "AUS", "USA", "CAN", and "SGP" were more likely to be the "migrant colonies", while the right-hand-side countries such as "MLI", "NGA", "MDG", and "GHA" were more likely to be the "extractive states".

(b) The linear-log model can be written as follows:

$$Protection_i = \beta_0 + \beta_1 log(Mortality_i) + u_i, \ i = 1, ..., n, \tag{9}$$

where $Protection_i$ denotes the index of "Protection Against Expropriation Risk" (1985-95) for the i-th former European colony, and $Mortality_i$ denotes the "Settler Mortality" (19th century) for the i-th colony. Given the definition of the linear-log model,

$$\beta_1 \approx \frac{\Delta Protection}{\Delta Mortality/Mortality}. \tag{10}$$

Therefore, supposing that $Mortality$ changes by 1 percent, we have

$$\Delta \widehat{Protection} \approx \hat{\beta}_1 \cdot \frac{\Delta Mortality}{Mortality} = -0.414 \times 1\% = -0.00414 \text{ (unit of } Protection). \tag{11}$$

5. (a) STATA commands:

reg salary top10 r11_25 r26_40 r41_60 GPA llibvol lcost

The estimated OLS coefficients are 3284.98 ($\hat{\beta}_6$) and 1779.92 ($\hat{\beta}_7$) for *llibvol* and *lcost*, respectively. Both have the representation of a linear-log model, that is,

$$\begin{aligned}\Delta \widehat{Salary} &\approx \hat{\beta}_6 \cdot \frac{\Delta libvol}{libvol} = 3284.96 \times 1\% = 32.85 \text{ (USD)},\\ \Delta \widehat{Salary} &\approx \hat{\beta}_7 \cdot \frac{\Delta cost}{cost} = 1779.92 \times 1\% = 17.80 \text{ (USD)}.\end{aligned}$$

```
. reg salary top10 r11_25 r26_40 r41_60 GPA llibvol lcost
```

Source	SS	df	MS			
				Number of obs	=	136
				F(7, 128)	=	159.22
Model	1.8344e+10	7	2.6206e+09	Prob > F	=	0.0000
Residual	2.1067e+09	128	16458377	R-squared	=	0.8970
				Adj R-squared	=	0.8914
Total	2.0451e+10	135	151485812	Root MSE	=	4056.9

salary	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
top10	27756.33	2218.722	12.51	0.000	23366.21	32146.45
r11_25	21724.47	1621.093	13.40	0.000	18516.86	24932.08
r26_40	11070.08	1440.387	7.69	0.000	8220.027	13920.13
r41_60	6456.605	1174.121	5.50	0.000	4133.406	8779.805
GPA	7382.992	2762.113	2.67	0.008	1917.679	12848.31
llibvol	3284.982	1176.495	2.79	0.006	957.0853	5612.879
lcost	1779.923	1071.077	1.66	0.099	-339.3859	3899.232
_cons	-26686.04	15308.94	-1.74	0.084	-56977.4	3605.31

(b) STATA commands:

reg salary top10 r11_25 r26_40 r41_60 GPA llibvol lcost

test top10 r11_25 r26_40 r41_60.

The homoskedasticity-only F-statistic is equal to 53.13. According to the critical value tables, the 5% critical value for the $\chi_4^2/4$ ($F_{4,\infty}$) is equal to 2.37. So the null hypothesis is rejected at the 5% level. In addition, we can see from the STATA result below, the p-value of the F-test is zero.

```
. test top10 r11_25 r26_40 r41_60

 ( 1)  top10 = 0
 ( 2)  r11_25 = 0
 ( 3)  r26_40 = 0
 ( 4)  r41_60 = 0

       F(  4,   128) =   53.13
            Prob > F =    0.0000
```

reg salary top10 r11_25 r26_40 r41_60 GPA llibvol lcost, robust

test top10 r11_25 r26_40 r41_60.

The heteroskedasticity-robust F-statistic is equal to 51.37, and the 5% critical value is the same as that for the homoskedasticity-only F-test. Therefore, we also reject the null hypothesis at the 5% level.

```
. test top10 r11_25 r26_40 r41_60

 ( 1)  top10 = 0
 ( 2)  r11_25 = 0
 ( 3)  r26_40 = 0
 ( 4)  r41_60 = 0

       F(  4,   128) =   51.37
            Prob > F =    0.0000
```